

Danmarks Tekniske Universitet

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Course title: Bayesian Machine Learning

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Aids allowed: All aids except internet

Exam duration: 4 hours

Weighting: 100%

02477 Bayesian Machine Learning Exam 2025R

Technical University of Denmark

- **Duration:** 4 hours
- **Aids:** All aids except internet
- **Student number:** Make sure your student number is visible on all pages.
- **Results:** Report all numeric results with 2 digits after the decimal point.
- **Explain** how you arrived at your results, document intermediate results when possible.
- **Hand-in:** Your solution must be handed in digitally as a PDF.

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Part 1: Linear Gaussian Systems

Consider the following linear system

$$\mathbf{z} = \mathbf{A}\mathbf{x} + \mathbf{b} + \mathbf{n}_1, \quad (1)$$

$$\mathbf{y} = \mathbf{W}\mathbf{z} + \mathbf{n}_2, \quad (2)$$

where $\mathbf{x}, \mathbf{z}, \mathbf{b}, \mathbf{y}, \mathbf{n}_1, \mathbf{n}_2 \in \mathbb{R}^2$ and

$$\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (3)$$

$$\mathbf{n}_1 \sim \mathcal{N}(\mathbf{0}, \frac{1}{2}\mathbf{I}) \quad (4)$$

$$\mathbf{n}_2 \sim \mathcal{N}(\mathbf{0}, \frac{1}{2}\mathbf{I}) \quad (5)$$

and the following constants are given by

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{W} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}. \quad (6)$$

Question 1.1: Determine the distribution $p(\mathbf{z}|\mathbf{x})$.

Question 1.2: Determine the distribution $p(\mathbf{x}|\mathbf{z})$.

Question 1.3: Determine the distribution $p(\mathbf{y})$.

Part 2: Binary classification

Consider the following supervised model

$$y_n | \mathbf{x}_n, \mathbf{w} \sim \text{Ber}(y | \sigma(\mathbf{w}^T \mathbf{x}_n)), \quad (7)$$

$$\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \lambda^{-1} \mathbf{I}), \quad (8)$$

where $\mathbf{x}_n \in \mathbb{R}^2$, $y_n \in \{0, 1\}$, $\mathbf{w} = [w_1 \ w_2]^T \in \mathbb{R}^2$, $\lambda = \frac{1}{4}$, $\sigma(\cdot)$ is the logistic sigmoid function and $\mathbf{I}$ is an identity matrix.

In Table 1 you are given a small dataset $\mathcal{D} = \{(\mathbf{x}_n, y_n)\}_{n=1}^N$ with $N = 2$ observations.

n	$\mathbf{x}_n$	y_n
1	$\begin{bmatrix} 0 & 0 \end{bmatrix}^T$	0
2	$\begin{bmatrix} 1 & 1 \end{bmatrix}^T$	1

Table 1: Dataset of $N = 2$ observations.

Let $\mathbf{y} = [0 \ 1]^T$ denote all the targets such that the n 'th element in $\mathbf{y}$ is y_n . Moreover, assume $\hat{\mathbf{w}}_{\text{MAP}}$ is a MAP estimator for the model such that

$$\hat{\mathbf{w}}_{\text{MAP}} = \arg \max_{\mathbf{w}} p(\mathbf{w} | \mathbf{y}) = 0.74077439 \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \quad (9)$$

Finally, let $\mathbf{x}^* = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

Question 2.1: Determine the posterior predictive distribution $p(y^* | \mathbf{y}, \mathbf{x}^*)$ using the plug-in approximation based on the MAP-estimator $\hat{\mathbf{w}}_{\text{MAP}}$.

Question 2.2: Evaluate $\log p(\mathbf{y}, \mathbf{w})$ at $\mathbf{w} = \hat{\mathbf{w}}_{\text{MAP}}$ for the given dataset.

Question 2.3: Determine the gradient and Hessian of $\log p(\mathbf{y}, \mathbf{w})$ with respect to $\mathbf{w}$ at $\mathbf{w} = \hat{\mathbf{w}}_{\text{MAP}}$.

If you did not solve the Question 2.3, you can assume the following Hessian matrix $\mathcal{H}$ when solving the next two questions.

$$\mathcal{H} = \frac{2}{5} \mathbf{I}.$$

Question 2.4: Determine the posterior predictive distribution $p(y^* | \mathbf{y}, \mathbf{x}^*)$ using the Laplace approximation.

Question 2.5: Suppose another target variable for regression, $z_n \in \{0, 1, 2, \dots\}$, is collected for each $\mathbf{x}_n$ such that $\mathcal{D}_2 = \{(\mathbf{x}_n, z_n)\}_{n=1}^N$ is another dataset for supervised learning. Explain how you would adapt the model in eq. (7)-(8) for the purpose of predicting z^* from $\mathbf{x}^*$ using the dataset $\mathcal{D}_2$.

Part 3: A simple mixture model

Consider the following probabilistic model for a single observation $y \in \mathbb{R}$ with parameters $s \in \{0, 1\}$ and $v \in \mathbb{R}$:

$$p(y|v, s) = \mathcal{N}(y|v, \sigma^2 + s\tau^2), \quad (10)$$

$$p(v) = \mathcal{N}(v|1, 1), \quad (11)$$

$$p(s) = \text{Ber}(s|p), \quad (12)$$

where $\sigma^2 = 1$, $\tau^2 = 2$ and $p = \frac{1}{4}$.

Question 3.1: Show that $p(y|v)$ is a Gaussian mixture model and determine the mixture weight, mean, and variance for each component.

Question 3.2: Determine the distribution $p(y)$ analytically.

Question 3.3: Determine the posterior probability $p(s = 1|y)$.

Part 4: Regression models

Consider the following probabilistic model:

$$p(y_n|\mathbf{w}, x_n) = \mathcal{N}(y_n|w_1 + w_2x_n, \sigma^2), \quad (13)$$

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I}), \quad (14)$$

where $\sigma^2 = \sqrt{2}$, $\alpha = \frac{1}{2}$, $\mathbf{w} = [w_1 \ w_2]^T \in \mathbb{R}^2$ is the parameters of the model, $x_n \in \mathbb{R}$ is an input feature and $y_n \in \mathbb{R}$ is the corresponding target variable for the n 'th observation. Suppose a dataset of N observations has been collected (but not given here) and gave rise to the following posterior distribution

$$p(\mathbf{w}|\mathbf{y}) = \mathcal{N}(\mathbf{w}|\mathbf{m}, \mathbf{S}), \quad (15)$$

where $\mathbf{y}$ denotes the observed targets and

$$\mathbf{m} = \begin{bmatrix} -0.87 \\ 2.13 \end{bmatrix}, \quad \mathbf{S} = \begin{bmatrix} 0.41 & 0.11 \\ 0.11 & 0.27 \end{bmatrix}. \quad (16)$$

Question 4.1: Determine prior and posterior probability of the event $w_1 > 0$.

Question 4.2: Determine a 80% credibility interval for $p(w_2|\mathbf{y})$.

Question 4.3: Determine the posterior distribution $p(f^*|\mathbf{y}, x^*)$ for $f^* = w_1 + w_2x^*$ and $x^* = 1$.

Question 4.4: Determine the posterior predictive distribution $p(y^*|\mathbf{y}, x^*)$ for $x^* = 1$.

Suppose the model in eq. (13)-(14) does not fit the data well, and therefore, the model is extended to be able to capture heteroscedastic noise as follows

$$p(y_n|\mathbf{w}, x_n) = \mathcal{N}(y_n|w_1 + w_2x_n, e^{w_3 + w_4x_n}), \quad (17)$$

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I}), \quad (18)$$

where $\mathbf{w} = [w_1 \ w_2 \ w_3 \ w_4]^T \in \mathbb{R}^4$ is the new parameter vector to be estimated. Furthermore, let $q(\mathbf{w})$ be a variational approximation of the posterior distribution of the heteroscedastic model, i.e. $p(\mathbf{w}|\mathbf{y}) \approx q(\mathbf{w})$, where

$$q(\mathbf{w}) = \mathcal{N}(w_1|\mu_1, v_1)\mathcal{N}(w_2|\mu_2, v_2)\mathcal{N}(w_3|\mu_3, v_3)\mathcal{N}(w_4|\mu_4, v_4)$$

with the following means and variances:

$$\boldsymbol{\mu} = [-0.82 \ 2.37 \ -0.88 \ 0.37]^T$$
$$\mathbf{v} = [0.38 \ 0.19 \ 0.13 \ 0.09]^T,$$

such that the i 'th element in $\boldsymbol{\mu}$ is μ_i etc.

Question 4.5: Determine the entropy of $q(\mathbf{w})$.

Question 4.6: Evaluate the (approximate) posterior predictive density $p(y^*|\mathbf{y}, x^*)$ for $y^* = 1$ and $x^* = 1$ using the variational approximation $q(\mathbf{w})$ and Monte Carlo estimation.

Part 5: A non-linear model

Consider the following non-linear model

$$f(x_n) = w_2 \tanh(w_1 x_n) \quad (19)$$

with two parameters $\mathbf{w} = [w_1 \ w_2]^T \in \mathbb{R}^2$ and further assume

$$y_n = f(x_n) + \epsilon, \quad (20)$$

where $\epsilon \sim \mathcal{N}(0, \sigma^2)$ is i.i.d. Gaussian noise with variance $\sigma^2 = 2$. Finally, assume a Gaussian prior distribution for $\mathbf{w}$, i.e., $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

Let $p(\mathbf{w}|y_1)$ denote the posterior distribution after observing a single observation with $x_1 = 1$ and $y_1 = 2$.

Question 5.1: Run a single MCMC chain using the Metropolis algorithm to generate samples from the posterior distribution $p(\mathbf{w}|y_1)$. Use 10^4 iterations with a standardized Gaussian as proposal distribution. Initialize the chain at $\mathbf{w} = [0 \ 0]^T$. Discard 10% of the samples as warm up. Plot the resulting traces for both parameters.

Hint: The hyperbolic tangent function ($\tanh$) can be evaluated in Jax using `jnp.tanh` or in numpy using `np.tanh`.

Use the posterior samples of $\mathbf{w}$ from Question 5.1 to answer the next two questions. If you did not solve the previous question, you can draw 10^4 samples from the prior distribution of $\mathbf{w}$, i.e.

$$\mathbf{w} \sim \mathcal{N}(\mathbf{w}|\mathbf{0}, \mathbf{I}). \quad (21)$$

Question 5.2: Estimate the posterior probability of $f(x^*) > 1$ for $x^* = 2$.

Question 5.3: Estimate the mean and a 95% interval for the posterior predictive distribution $p(y^*|y_1, x^*)$ for $x^* = 2$.